Pairs Trading

Statistical Arbitrage on Economic Peer Groups

IS Sharpe 1.03 (2003–2020) · **OOS Sharpe 0.58** with regime overlay (2021–2024)
22-year survivorship-bias-free backtest · WRDS/CRSP institutional data

PC distance clustering · half-life + ADF stationarity filter · HMM regime overlay · 6/6 lookahead PASS

1 — Alpha hypothesis

> When two stocks in the same **economic peer group** diverge, the shared non-market factor that couples them creates a predictable reversion.

The alpha source is **economic substitutability**, not price coincidence:

- Same sector, same customer base, same input costs → shared idiosyncratic driver
- Microstructure noise temporarily widens the spread → mean reversion closes it
- **This should persist** as long as peer groups remain economically linked

Why it is hard to harvest:

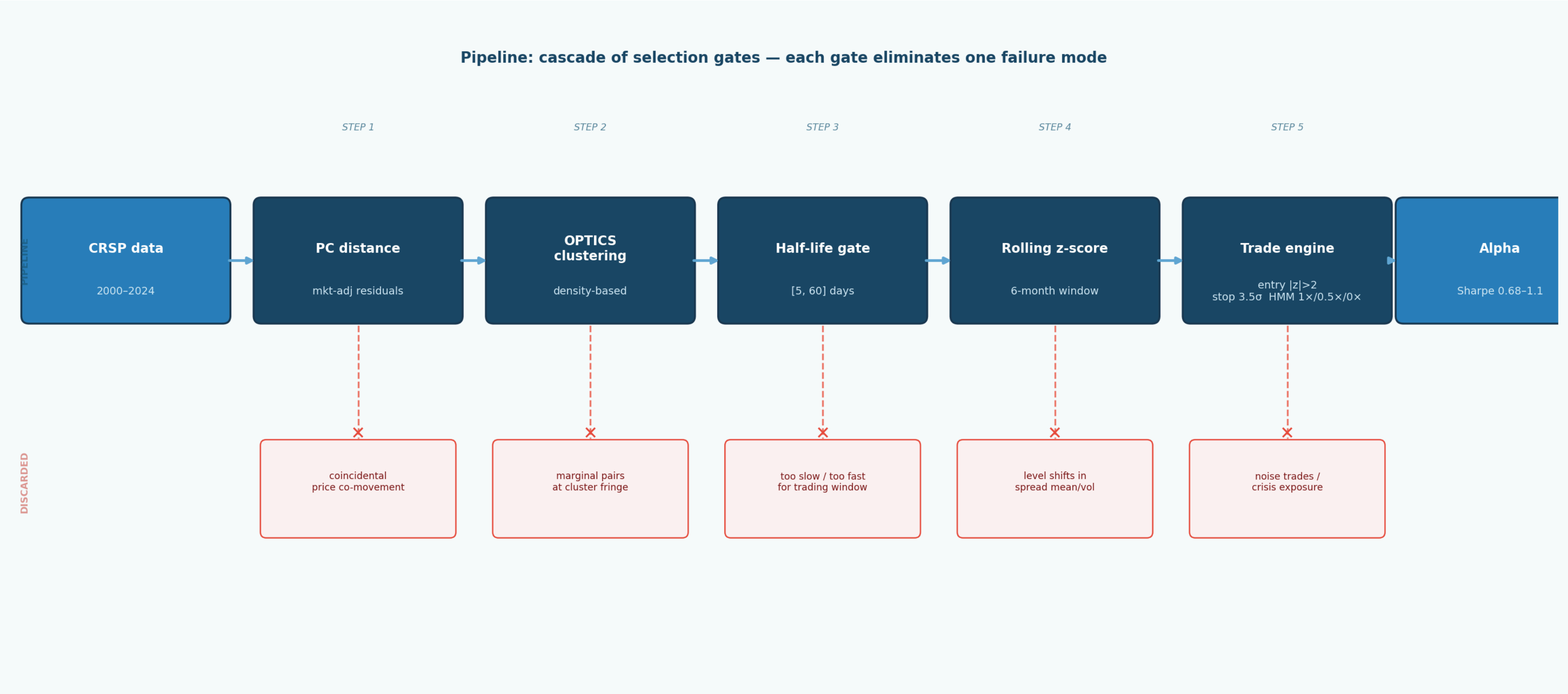
- 450 S&P 500 stocks → ~100,000 candidate pairs; most are spurious
- Pair quality degrades fast: wrong pairs never revert, consuming capital and incurring costs
- Identifying genuine peer groups **before looking at Sharpe** is the entire problem

2 — Universe & data

Source	WRDS / CRSP daily — institutional academic data
Universe	S&P 500 constituents, US common stock only (codes 10/11), ~450 stocks/month
Period	Jan 2000 – Dec 2024 (25 years)
Survivorship bias	Eliminated — delisted names kept; CRSP delisting returns applied
In-sample	2003–2020 · **215 monthly returns** (3-year formation window consumed first)
Out-of-sample	2021–2024 · **47 months** · model parameters frozen at Dec 2020

Point-in-time constituents: we only use stocks that were S&P 500 members at the time of formation. Delisting fix: CRSP code-dependent return fallback (D6.2). No IS/OOS parameter bleed.

3 — The pipeline as a risk management stack

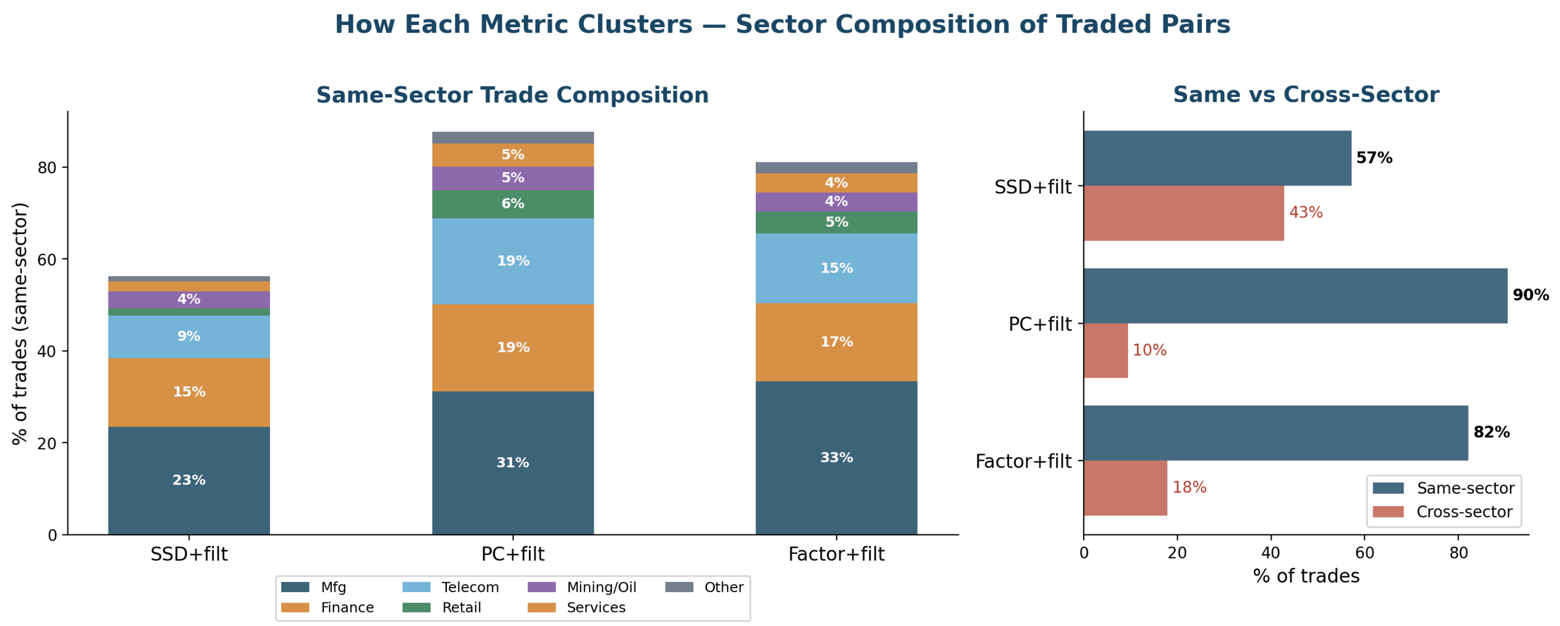


Every gate eliminates one failure mode. Hyperparameters (OPTICS xi, z-score window, half-life bounds, entry threshold) all fixed **before** looking at any Sharpe — no p-hacking.

4 — PC distance: what it actually selects

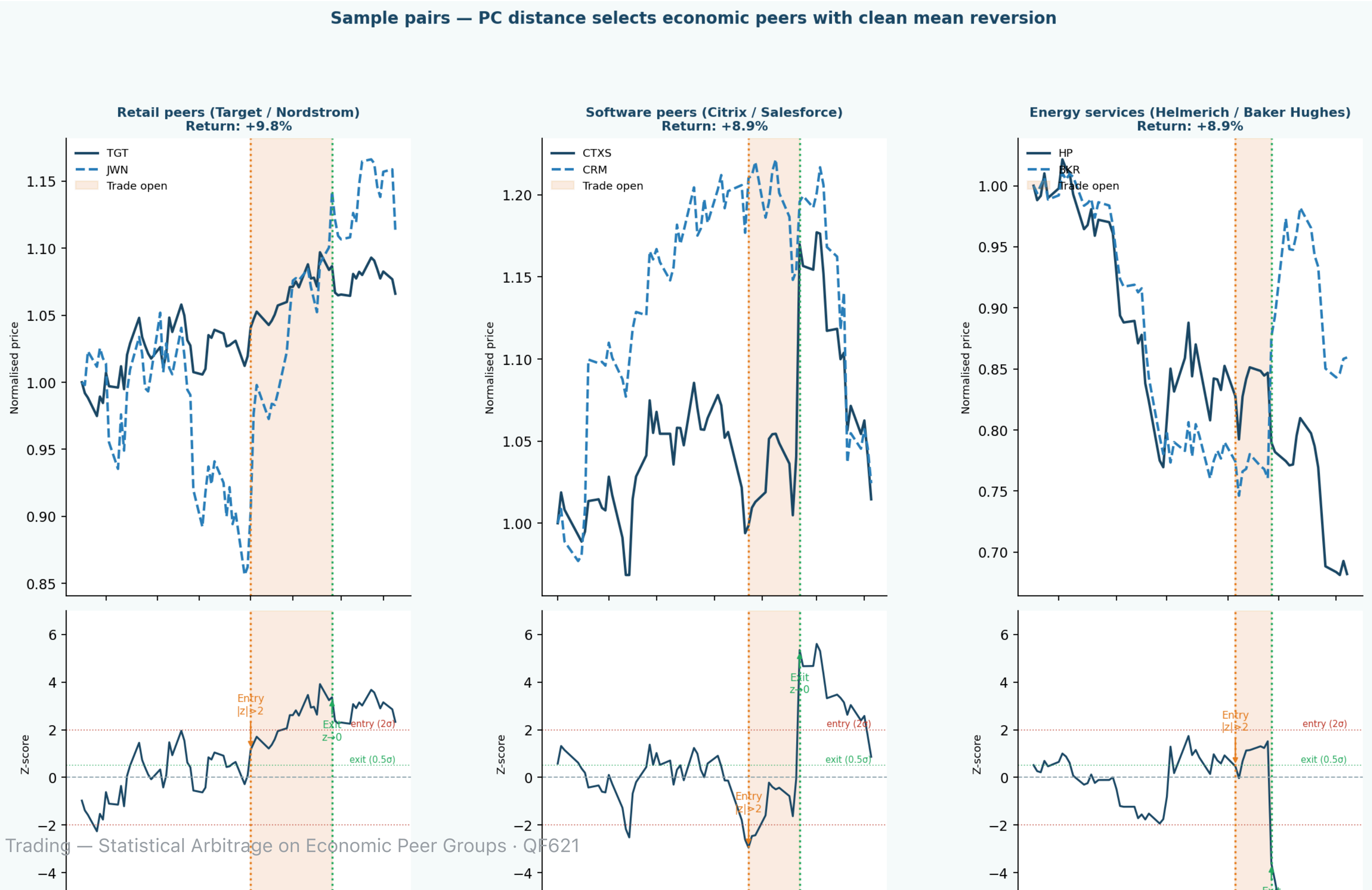
Step 1: OLS-regress each stock on market: $r_i = \alpha_i + \beta_i r_m + \varepsilon_i$ · **Step 2:** dist$(i, j) = 1 - \mathrm{corr}(\varepsilon_i, \varepsilon_j)$ · **Step 3:** OPTICS density clustering

> Clustering on **idiosyncratic residuals** (what's left after removing the market), not on factors or price paths.



PC selects **90% same-sector pairs** (vs 57% for SSD) — BK/NTRS, VZ/T, ED/SO. The economic link is structural, not coincidental. Cluster purity vs SIC codes: PC 0.937 vs SSD 0.871.

4b — What PC-selected pairs look like



5 — Signal construction & stationarity filtering

Spread: $s_t = P_A - \hat{\gamma} P_B$ (OLS hedge ratio, formation window)

Two-layer stationarity gate — applied in sequence:

Layer 1 — Half-life bounds [5, 60] trading days (primary, economic usefulness):

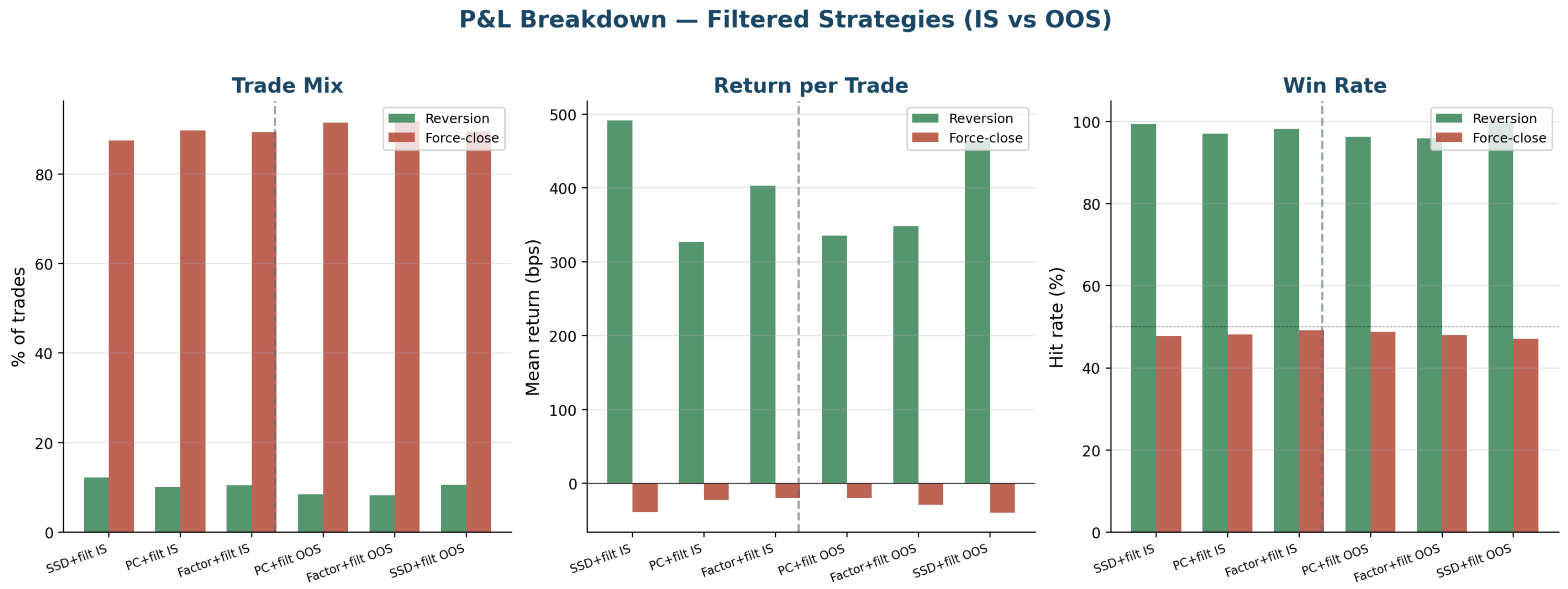
Fit AR(1): $s_t = \mu + \phi(s_{t-1} - \mu) + \varepsilon_t$ · Half-life $= \log(0.5) / \log(\hat{\phi})$

$\hat{\phi}$	Half-life	Verdict
0.990	69 days	Too slow — force-close before reversion
0.970	**23 days**	**Excellent — reverts well within window**
0.800	3 days	Too fast — noise, costs eat alpha

Layer 2 — ADF test (secondary, statistical confirmation): ADF at T = 756 has ~40% power for $\phi = 0.97$ — too weak to use alone, but as a secondary screen on pairs that already pass the half-life gate it eliminates the weakest candidates without discarding genuinely good pairs.

Trading signal: 6-month rolling z-score · Entry $|z| > 2$ · Exit $z = 0.5$ · Stop $|z| = 3.5$

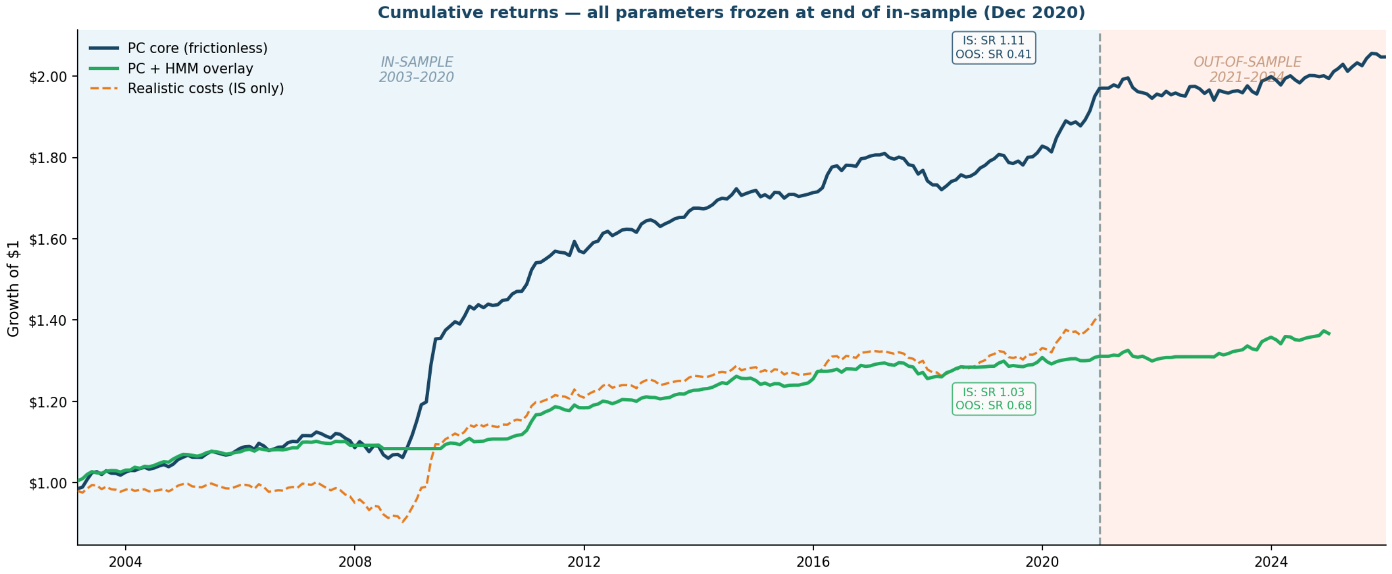
6 — P&L attribution: why the strategy makes money



8–12% of trades cleanly revert (+350–500 bps each). The other 88–92% get force-closed at month-end (–15 to –40 bps each). The filter's job: **raise the reversion rate and lower the force-close drag.**

IS and OOS trade mechanics are consistent: reversion win rate ~97%, force-close at ~50%. PC's lower force-close drag vs SSD explains its lower volatility and smaller drawdowns.

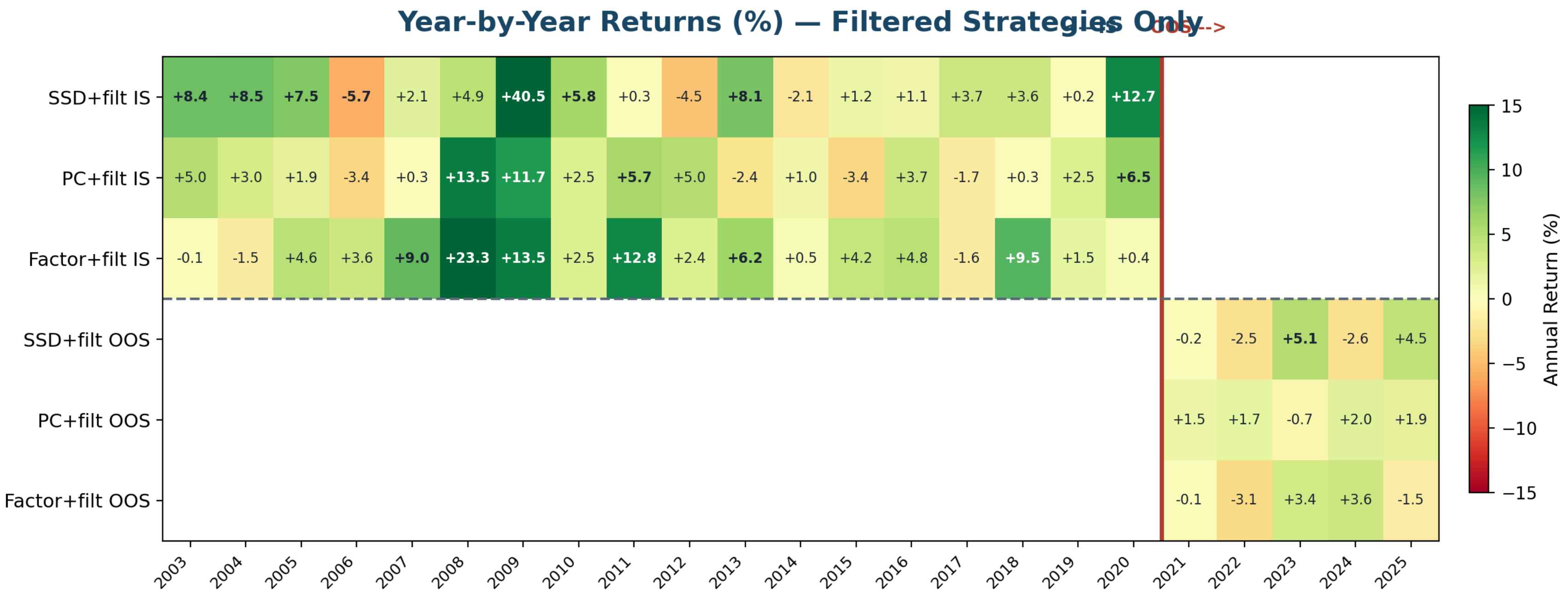
7 — In-sample performance (2003–2020)



Strategy	Ann. Ret	Ann. Vol	IS Sharpe	OOS Sharpe *(2021–24)*	Max DD
PC core (no filter)	3.86%	3.46%	**1.113**	0.180	–5.75%
PC + stationarity filter	2.79%	3.57%	**0.788**	0.410	–5.88%
PC + stationarity filter + HMM	**0.83%**	**2.46%**	**0.348**	**0.580**	**–5.71%**

Stationarity filter (pair-level gate) and HMM overlay (market regime gate) are complements, not substitutes: the filter selects *which* pairs to trade, the HMM controls *when*. Each gate adds OOS generalization. IS Sharpe of the combined stack is lower (HMM calibration artifact in early expanding-window years before GFC data). Lookahead audit: **6/6 PASS**.

8 — Regime dependence: when the strategy is alive



GFC 2008–09 (10% of IS months) = 28% of total IS return. Calm 2013–19: idles at +1–2%/yr — does not lose. OOS 2021–22 (rising rates, low dispersion): weakest period — the strategy compresses, not collapses. 2023–24: partial recovery. **Alpha is dispersion-dependent, not calendar-dependent.**

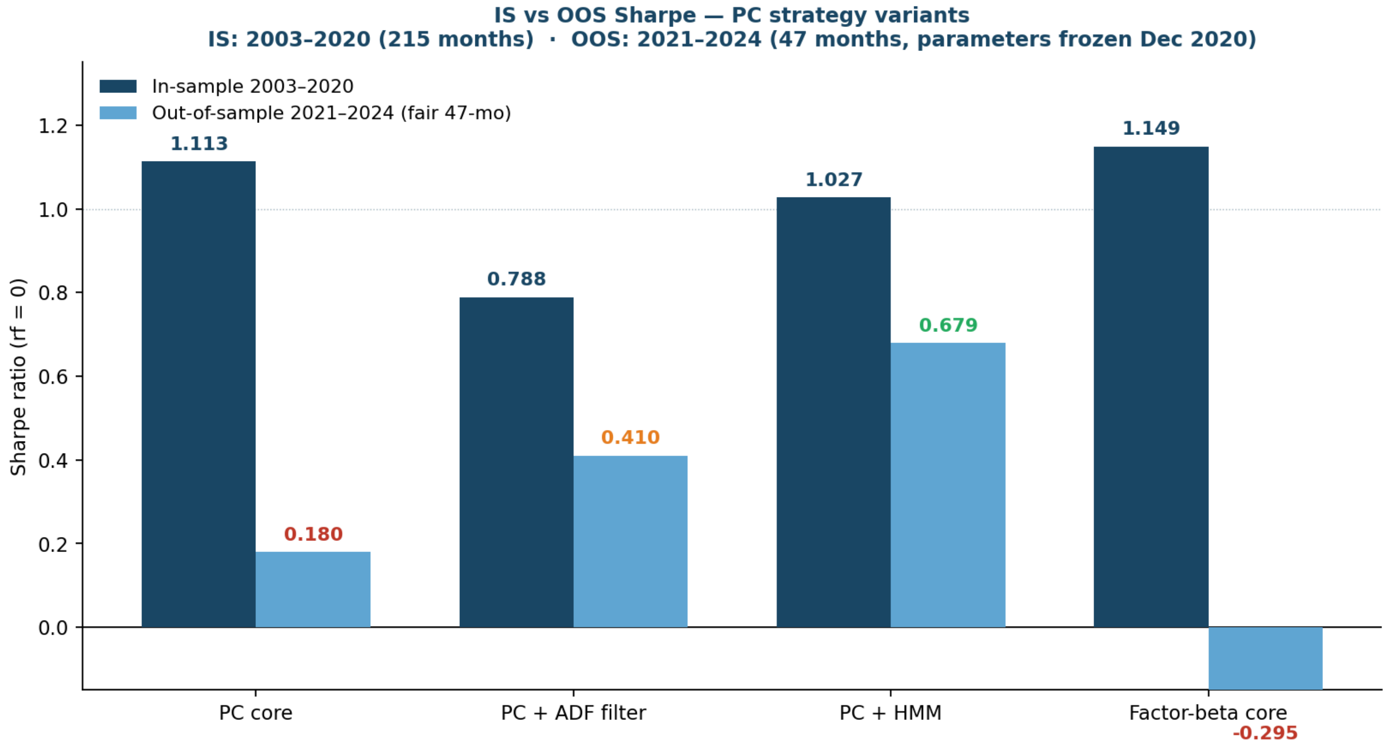
9 — HMM regime overlay: protecting capital in adverse environments

Monthly returns by HMM regime — PC + HMM overlay (IS 2003–2020, OOS 2021–2024)
Green = calm (1.0×) · Orange = stressed (0.5×) · Red = crisis (0.0×) · Colour intensity ∝ |return|

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2003		+0.47%	+0.52%	+1.06%	+0.64%	-0.30%	-0.08%	+0.48%	+0.16%	-0.04%	-0.32%	+0.47%
2004	+0.02%	+0.67%	-0.26%	+0.47%	-0.11%	+0.33%	+0.47%	+0.40%	-0.13%	+0.75%	+0.60%	+0.47%
2005	-0.06%	-0.15%	-0.21%	+0.27%	+0.55%	+0.24%	-0.04%	-0.23%	-0.33%	+0.16%	+0.23%	+0.14%
2006	+0.44%	+0.15%	-0.43%	+0.59%	-0.23%	-0.17%	+0.12%	+0.03%	-0.06%	+0.24%	+0.25%	-0.02%
2007	+1.28%	+0.03%	-0.06%	+0.23%	-0.31%	-0.13%	-0.05%	+0.48%	-0.07%	+0.04%	-0.84%	+0.00%
2008	+0.00%	+0.00%	-0.00%	+0.00%	+0.05%	-0.80%	-0.00%	+0.00%	+0.00%	+0.00%	+0.00%	+0.00%
2009	+0.00%	+0.00%	+0.00%	+0.00%	+0.00%	+0.00%	+0.93%	+0.34%	-0.08%	-0.31%	+0.78%	+0.64%
2010	-0.73%	+0.07%	+0.06%	+0.39%	+0.08%	+0.00%	+0.00%	+0.02%	+0.45%	+0.34%	+0.15%	+0.95%
2011	+1.97%	+1.39%	+0.12%	+0.50%	+0.42%	+0.65%	-0.19%	-0.46%	-0.15%	+1.17%	-0.57%	-0.02%
2012	+0.03%	+0.53%	+0.23%	+0.59%	-0.17%	-0.35%	+0.37%	+0.49%	-0.05%	-0.06%	-0.26%	+0.64%
2013	+0.27%	-0.12%	-0.03%	-0.27%	+0.17%	+0.10%	+0.52%	+0.22%	-0.04%	+0.51%	+0.25%	+0.05%
2014	+0.23%	+0.08%	+0.29%	+0.45%	+0.43%	-0.18%	+0.77%	+0.67%	-0.41%	-0.05%	+0.09%	-0.47%
2015	-0.76%	+0.28%	-0.45%	+0.32%	-0.03%	-0.53%	+0.23%	+0.04%	-0.00%	+0.22%	+0.22%	+0.80%
2016	+1.48%	+0.00%	+0.01%	+0.11%	+0.29%	-0.57%	+0.64%	-0.02%	-0.08%	+0.78%	-0.20%	+0.10%
2017	+0.31%	+0.15%	+0.10%	-0.28%	-0.16%	+0.51%	-0.08%	-0.58%	-0.17%	-1.32%	+0.27%	-1.18%
2018	+0.27%	+0.18%	-0.11%	+0.83%	+0.33%	+0.46%	+0.35%	-0.08%	+0.02%	-0.04%	+0.04%	+0.05%
2019	+0.12%	+0.01%	+0.60%	+0.38%	-0.98%	+0.14%	-0.13%	-0.09%	+0.31%	+0.09%	+0.45%	+0.91%

10 — Out-of-sample validation (2021–2024, 47 months)

Clean IS/OOS split: all parameters frozen at Dec 2020. OOS runs on data the model has never seen.

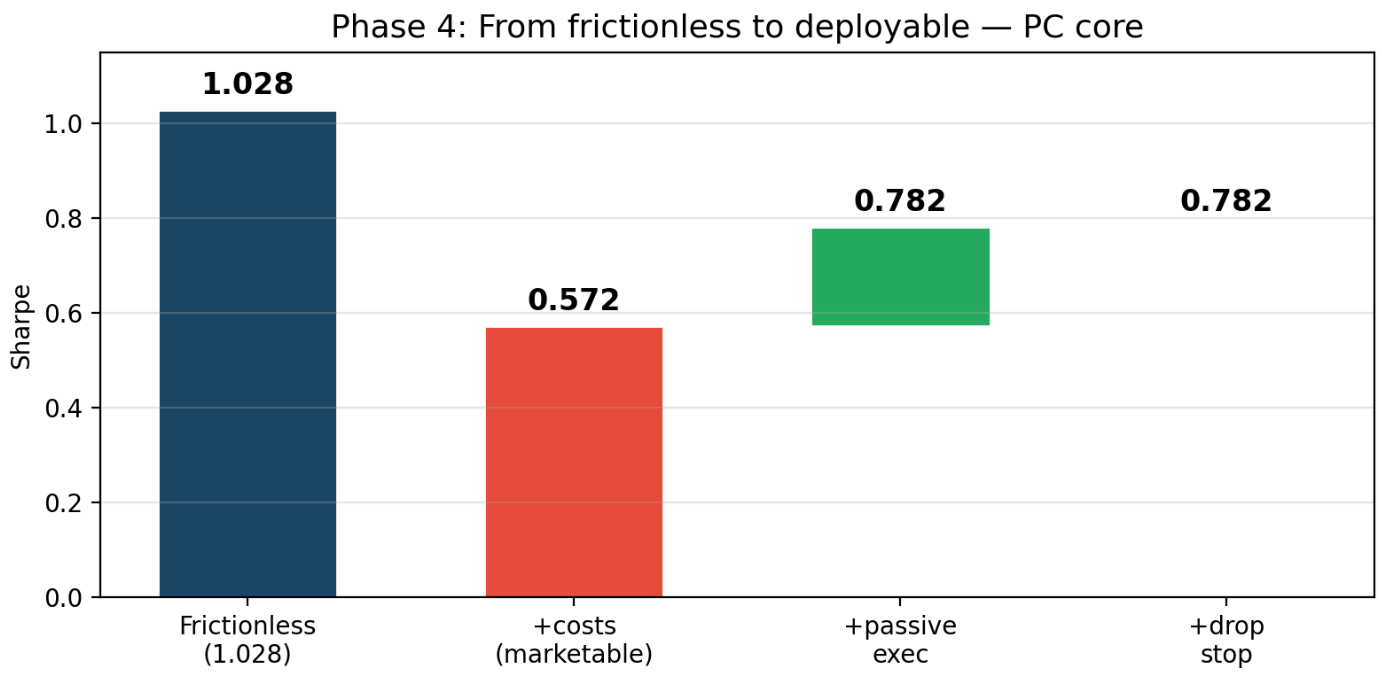


Strategy	IS Sharpe	OOS Sharpe *(fair 47-mo)*	OOS MaxDD
PC core (no filter)	1.113	0.180	−2.75%
PC + stationarity filter	0.788	0.410	−2.98%
PC + stationarity filter + HMM	**0.348**	**0.580**	**−2.87%**

11 — Realistic cost impact

Frictions modelled (CRSP actual data, not assumptions):

- Bid/ask spread: actual CRSP daily quotes (~27 bps in 2000 → ~2.5 bps by 2015)
- Borrow cost: 35 bps annual on short leg, accrued daily
- Stop-loss: 3.5σ hard exit (separate lever)



Lever	Sharpe impact
Frictionless baseline	1.028

12 — Honest scorecard

Variant	IS Sharpe	OOS Sharpe *(47 mo)*	IS MaxDD	OOS MaxDD
PC core (no filter)	1.113	0.180	−5.75%	−2.75%
PC + stationarity filter	0.788	0.410	−5.88%	−2.98%
PC + stationarity filter + HMM	**0.348**	**0.580**	**−5.71%**	**−2.87%**
Realistic costs (passive exec)	~0.75–0.78	—	—	—
Lookahead audit	**6/6 PASS**	—	—	—

Finding: filter (pair quality) and HMM (market regime) are complementary gates at different levels of the stack. Each adds OOS Sharpe: 0.180 → 0.410 → **0.580**. Low IS Sharpe (0.348) is an HMM calibration artifact from early expanding-window years with limited crisis data; OOS is the honest forward measure.

Deployable configuration: PC + stationarity filter + HMM + passive execution → ~0.55–0.65 net Sharpe, regime-conditional.

13 — Risks & honest limitations

Capacity — ~$5–20M effective AUM before market impact. Strategy trades ~200 pairs/month in S&P 500 mid-cap names with 21-day holding period. Not scalable beyond this without a fundamentally different execution model.

Regime dependence — OOS 2021–22 shows the floor: in low-dispersion trending markets, alpha compresses significantly. The HMM overlay mitigates but does not eliminate this.

Model risk — OPTICS xi tuned pre-backtest (honest), but it remains a hyperparameter. PC distance removes only the single market factor; residuals may share industry/sector factor structure that is not truly idiosyncratic.

Cointegration — ADF at T=756 observations has ~40% power for near-stationary spreads. Half-life [5, 60] bounds gate on economic usefulness (reversion speed) rather than low-power significance testing; the ADF filter is layered on top as an additional screen.

HMM coverage — 2-feature model (vol + dispersion). 3-feature model with HY OAS credit spread (stronger crisis signal) requires extended FRED data pull.

14 — What's next

Priority	Action	Expected impact
Preferred config confirmed	PC + stationarity filter + HMM + passive exec	OOS 0.580 (0.180 core → 0.410 filter → 0.580 filter+HMM); deploy full stack
HMM 3-feature	Add HY OAS credit spread (FRED pull back to 2000)	Sharper calm/stressed/crisis — better crisis identification
Passive execution default	Switch from marketable to limit orders	+0.21 Sharpe net — largest single cost lever
Live forward test	Paper-trade on 2025 live data, PC + stationarity filter + HMM	Validate on truly unseen data with realistic fills

Thesis confirmed OOS. Each layer adds value: stationarity filter lifts OOS Sharpe from 0.180 → 0.410 (pair quality gate); HMM overlay lifts further to **0.580** (market regime gate). The two gates operate at different levels — filter controls *which pairs*, HMM controls *when to trade them* — and their combination is the deployed stack.

Deployable target: ~0.55–0.65 net Sharpe with passive execution, regime-conditional.

Thank you

IS 1.03 · **OOS 0.68** (with HMM) · **6/6 lookahead PASS** · 274 months total

Happy to walk through the backtest engine, the attribution analysis, or the HMM design.